

Program : Diploma in Instrumentation Engineering	
Course Code : 6082D	Course Title: Mechatronics and Control
Semester : 6	Credits: 4
Course Category: Program Elective	
Periods per week : 4 (L:3 T:1 P:0)	Periods per semester: 60

Course Objectives:

- To acquire knowledge on mechatronics, measurement systems and different actuators.
- To build PLC programs using ladder logic.
- To know about designing of mechatronics systems.

Course Prerequisites:

Topic	Course code	Course name	Semester
Knowledge of basic Mathematics		Engineering Mathematics I	I

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Describe about various types of sensors and transducers.	13	Understanding
CO2	Explain the various mechanical, electrical and pneumatic actuation systems.	14	Understanding
CO3	Build ladder diagrams for PLC programming.	18	Applying
CO4	Describe examples of mechatronics system.	13	Understanding
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3			3				
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Describe about various types of sensors and transducers.		
M1.01	Describe the importance of mechatronics	3	Understanding
M1.02	Explain systems and controllers	5	Understanding
M1.03	Explain different sensors.	5	Understanding
Contents: Sensors and Transducers: Introduction to Mechatronics: Mechatronics - Importance of Mechatronics - Systems - Measurement systems - Control systems and their types - Closed - loop control System - Automatic water level controller - Sequential controllers - washing machine Measurement System terminology - Displacement, Position & Proximity Sensors - Velocity and Motion Sensors - Force Sensors - Fluid Pressure Sensors - Flow Sensors - Liquid Level Sensors, Temperature.			
CO2	Explain the various mechanical, electrical and pneumatic actuationsystems		
M2.01	Explain mechanical actuation systems	4	Understanding
M2.02	Describe electrical actuation systems	5	Understanding
M2.03	Explain pneumatic actuation systems	5	Understanding
	Series Test - I	1	
Contents: Actuation systems: Mechanical Actuation Systems: Types of motion - Freedom and constraints - Loading - Gear Trains - Pawl & Ratchet-Belt & Chain drives - Bearings - Selection - Ball & Roller bearings - Mechanical aspects of motor selection. Electrical Actuation Systems - Switches & Relays, Solenoids - D.C Motors - A.CMotors - Stepper Motors - Specifications and Control of stepper motors - Servomotors - D.C Servomotor and A.C Servomotor. Pneumatic & Hydraulic Systems - Power supplies - DCV, PCV – Cylinders - Rotary actuators.			

CO3	Build ladder diagrams for PLC programming.		
M3.01	Explain block diagram and architecture of PLC	6	Understanding
M3.02	Design ladder diagrams	5	Applying
M3.03	Explain counter, data handling and shift registers.	4	Understanding
M3.04	Describe selection of PLC.	3	Understanding
Contents: Programmable Logic Controller: Programmable Logic Controller (PLC): Definition-Basic block diagram and structure of PLC-Input/Output processing- PLC Programming- Ladder diagram, its logic functions-Latching and Sequencing- PLC mnemonics-Timers-Internal relays and Counters- Shift registers- Master and Jump Controls- Data handling-Analog input/output- Selection of PLC-calculations.			
CO4	Describe examples of mechatronics system.		
M4.01	Explain design steps of mechatronics	6	Understanding
M4.02	Explain monitoring systems of mechatronics.	7	Understanding
	Series Test - II	1	
Contents: Mechatronics systems : Design Examples & Advanced Applications in Mechatronics - Design process stages - Traditional Vs. Mechatronics designs - Possible design solutions - Timed switch - Wind - screen wiper motion - Bath room scale - Case studies of Mechatronics systems - A pick - and - place robot - Car park barrier -Car engine management system - Automatic Camera and Automatic Washing Machine only - Sensors for Condition Monitoring Systems of Production Systems - Examples of Monitoring methods - Vibration monitoring - Temperature monitoring - Wear behavior monitoring - Mechatronics control in automated manufacturing - Monitoring of Manufacturing processes - On - line quality monitoring			

Text / Reference:

T/R	Book Title/Author
R1	W. Bolton, Pearson Education India - Mechatronics.
R3	R. K. Rajput, S. Chand & Co, New Delhi -.A Text Book on Mechatronics
R4	M. D. Singh& Joshi, Prentice Hall of India - Mechatronics.
R5	HMT, Tata McGraw Hill, New Delhi - Mechatronics.
R6	Devadas Shetty, PWS Publishing -Mechatronics System
R7	Pradeep KumarSrivatsava, BPB Publications-. Exploring Programmable Logic Controllers with applications

Online Resources:

Sl. No	Website Link
1	https://web.itu.edu.tr/yalcinme/files/courses/MMG/ch1%20introduction%20to%20mechatronics.pdf
2	https://www.meprogram.com.au/wp-content/uploads/2018/03/MFE3004AdditionalPT5.pdf
3	https://www.electrical4u.com/programmable-logic-controllers/
4	http://www.justscience.in/articles/applications-of-mechatronics/2017/07/11
5	https://www.engineering.unsw.edu.au/mechanical-engineering/study-with-us/types-of-mechanical-and-manufacturing-engineering/mechatronic-engineering